

A SHARP LARGEST-GAP INEQUALITY FOR $A, 2A, 3A$

GPT-5.5 PRO

ABSTRACT. For a finite set $A \subset \mathbb{R}$, write $g(A)$ for the largest gap between consecutive elements, and write $d(A) = \max A - \min A$. We prove the sharp inequality

$$g(A) + g(3A) - 2g(2A) \geq -\frac{d(A)}{12}.$$

The constant is best possible: the finite sets

$$\{0, \dots, m\} \cup \{3m, \dots, 4m\} \cup \{7m\} \cup \{10m, \dots, 12m\}$$

have normalized value $-1/12 + 1/(12m)$. The lower bound is proved by reducing any strict counterexample to a completed union of at most five intervals or points. The remaining finite-dimensional problem is eliminated by an exact rational Farkas certificate. The accompanying verifier reconstructs the linear systems from the definitions and checks the certificate using only rational arithmetic.

REMARKS ON AI USE

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At the time of writing, the proof has *not* been verified by any human, nor has it been formalized. I (Boon Suan Ho) personally believe that it is more or less correct, but this judgement is formed from my experiences with ChatGPT, rather than from me studying the paper.

Terence Tao recently [said](#) that the problem-solving component of mathematical research can roughly be decomposed into proof generation, proof verification, and proof digestion. I am only claiming proof generation here: I want to make it clear that I have not made any effort to verify or digest the proof. I am sharing this document because I believe that there may be people interested in reading and understanding it.

I have made the auxiliary files that accompany this paper available at https://boonsuan.github.io/misc/sumset_bound.zip. You can reach me at hbs@u.nus.edu if you have any comments.

1. INTRODUCTION

Let $A = \{a_1 < \dots < a_h\}$ be a finite set of real numbers, $h \geq 2$. Define

$$g(A) = \max_{1 \leq i < h} (a_{i+1} - a_i), \quad d(A) = a_h - a_1,$$

and let $nA = A + \dots + A$ denote the n -fold sumset. A question of Wiesner on MathOverflow asks for lower bounds on

$$\frac{g(A) + g(3A) - 2g(2A)}{d(A)}.$$

The example

$$\{0, 1, 2, 6, 10, 11, 14, 15\}$$

shows that the numerator can be negative, giving the value $-1/15$; the question asks, in particular, whether the lower bound is smaller than this value [1]. We prove that the exact infimum is $-1/12$.

It is convenient to prove a slightly more general compact-set statement. If $K \subset \mathbb{R}$ is nonempty compact, define

$$\text{gap}(K) = \sup\{y - x : x < y, x, y \in K, (x, y) \cap K = \emptyset\},$$

with the convention that the supremum of the empty set is 0. For a finite set this agrees with the largest gap between consecutive elements. If $\text{gap}(K) > 0$, this supremum is attained: for any $\varepsilon > 0$ there are only finitely many disjoint gaps of length at least ε in a bounded interval. Also write

$$\text{diam}(K) = \max K - \min K.$$

Our main theorem is the following.

Theorem 1.1. *For every nonempty compact set $K \subset \mathbb{R}$,*

$$\text{gap}(K) + \text{gap}(3K) - 2\text{gap}(2K) \geq -\frac{\text{diam}(K)}{12}.$$

Consequently, for every finite set $A \subset \mathbb{R}$ with $|A| \geq 2$,

$$\frac{g(A) + g(3A) - 2g(2A)}{d(A)} \geq -\frac{1}{12}.$$

The compact form is useful because the natural extremal object is

$$K_0 = [0, 1] \cup [3, 4] \cup \{7\} \cup [10, 12].$$

For this set

$$\text{gap}(K_0) = 3, \quad \text{gap}(2K_0) = 2, \quad \text{gap}(3K_0) = 0, \quad \text{diam}(K_0) = 12,$$

so equality holds in Theorem 1.1. For finite sets, the following discretizations show that the same constant is the exact infimum.

Proposition 1.2. *The constant $1/12$ in Theorem 1.1 is best possible for finite sets.*

Proof. For an integer $m \geq 1$, let

$$B_m = \{0, 1, \dots, m\} \cup \{3m, 3m+1, \dots, 4m\} \cup \{7m\} \cup \{10m, 10m+1, \dots, 12m\}.$$

Then $d(B_m) = 12m$. The gaps of B_m have maximum $3m$, so $g(B_m) = 3m$. Direct addition gives

$$\begin{aligned} 2B_m &= \{0, \dots, 2m\} \cup \{3m, \dots, 5m\} \cup \{6m, \dots, 8m\} \\ &\quad \cup \{10m, \dots, 16m\} \cup \{17m, \dots, 19m\} \cup \{20m, \dots, 24m\}, \end{aligned}$$

so $g(2B_m) = 2m$. The triple sums of the four displayed blocks contain the integer intervals corresponding to

$$\begin{aligned} &[0, 3], [3, 6], [6, 9], [9, 12], [10, 14], [13, 17], \\ &[16, 20], [20, 25], [23, 28], [27, 31], [30, 36] \end{aligned}$$

after multiplication by m ; these intervals cover $[0, 36]$. Hence $3B_m = \{0, 1, \dots, 36m\}$ and $g(3B_m) = 1$. Therefore

$$\frac{g(B_m) + g(3B_m) - 2g(2B_m)}{d(B_m)} = \frac{3m + 1 - 4m}{12m} = -\frac{1}{12} + \frac{1}{12m}.$$

Letting $m \rightarrow \infty$ proves sharpness. $\square$

The rest of the paper proves the lower bound. The proof has two parts. First, by a completion argument, every strict counterexample is replaced by a finite union of intervals and points whose component gaps are all at least the largest gap of $2K$. Then elementary endpoint estimates show that a strict counterexample has at most five components. Second, the remaining finite-dimensional problem is checked by an exact rational certificate. The certificate is not a numerical optimization result: it is a list of rational Farkas identities, each of which is verified exactly.

2. GAPS AND COMPLETIONS

We record an elementary fact about gaps of compact sets. A *gap* of a compact set $E \subset \mathbb{R}$ is an open interval (u, v) such that $u, v \in E$ and $(u, v) \cap E = \emptyset$. Its length is $v - u$. Thus $\text{gap}(E)$ is the maximum length of a gap when E has a gap, and is 0 otherwise.

Lemma 2.1 (Monotonicity under filling). *Let $E \subseteq F \subset \mathbb{R}$ be compact sets with the same minimum and maximum. Then*

$$\text{gap}(F) \leq \text{gap}(E).$$

Consequently, if $K \subseteq K'$ are compact sets with the same minimum and maximum, then

$$\text{gap}(nK') \leq \text{gap}(nK) \quad (n \geq 1).$$

Proof. Let (u, v) be a gap of F . Since $E \subseteq F$, the interval (u, v) is disjoint from E . It is contained in a connected component of $\mathbb{R} \setminus E$. Because E and F have the same endpoints, that component is a bounded gap (u', v') of E , and $v - u \leq v' - u' \leq \text{gap}(E)$. Taking the supremum over gaps of F

proves the first assertion. The second follows because $nK \subseteq nK'$ and the two sumsets have the same endpoints, namely $n \min K$ and $n \max K$. $\square$

Throughout the proof we normalize

$$\min K = 0, \quad \max K = D.$$

Put

$$a = \text{gap}(K), \quad b = \text{gap}(2K), \quad c = \text{gap}(3K).$$

The desired inequality is

$$a + c - 2b \geq -\frac{D}{12}.$$

Equivalently, if

$$\delta = 2b - a - c,$$

then it is enough to prove

$$D \geq 12\delta$$

whenever $\delta > 0$. If $\delta \leq 0$, the inequality is immediate.

Lemma 2.2 (Completion). *Let $K \subset \mathbb{R}$ be compact, and let $(p, p+b)$ be a largest gap of $2K$, where $b = \text{gap}(2K) > 0$. Define two points of K to be equivalent if they can be joined by a finite chain of points of K in which consecutive distances are $< b$. Let $C_1, \dots, C_r$ be the resulting equivalence classes, and put*

$$K^\sharp = \bigcup_{i=1}^r \text{conv}(C_i).$$

Then $K^\sharp$ is a finite union of closed intervals or points whose component gaps are at least b , and

$$\text{diam}(K^\sharp) = \text{diam}(K), \quad \text{gap}(K^\sharp) \leq \text{gap}(K), \quad \text{gap}(3K^\sharp) \leq \text{gap}(3K),$$

and

$$\text{gap}(2K^\sharp) = b.$$

Proof. Call the equivalence classes in the definition of $K^\sharp$ the b -clusters of K . Each b -cluster is closed in K : if x_n lies in one cluster and $x_n \rightarrow x \in K$, then $|x_n - x| < b$ for all large n , so x lies in the same cluster. Distinct b -clusters are separated by distance at least b ; since K is bounded, there are only finitely many of them. Hence $K^\sharp$ is a finite union of closed intervals or points, and its component gaps are at least b .

The set $K^\sharp$ has the same endpoints as K , so the diameter is unchanged. The inequalities for $\text{gap}(K^\sharp)$ and $\text{gap}(3K^\sharp)$ follow from Lemma 2.1.

It remains to show that the selected gap $(p, p+b)$ of $2K$ survives after completion.

Let C and C' be two b -clusters. We claim that all sums $x + x'$, with $x \in C$ and $x' \in C'$, lie on the same side of the gap $(p, p+b)$. Indeed, suppose there were sums $x_0 + x'_0 \leq p$ and $x_1 + x'_1 \geq p+b$. Join x_0 to x_1 inside C by steps of size $< b$, and join x'_0 to x'_1 inside C' by steps of size $< b$. Changing one

coordinate at a time gives a chain of sums in $2K$ starting at $\leq p$ and ending at $\geq p + b$, with every step of size $< b$. Since no term of the chain can lie in $(p, p + b)$, the first step that crosses from the left side to the right side has size at least b , a contradiction.

Thus $C + C'$ is entirely contained in $(-\infty, p]$ or entirely contained in $[p + b, \infty)$. Consequently

$$\text{conv}(C) + \text{conv}(C')$$

is also entirely contained on the same side, since it is the interval between the smallest and largest sums from $C + C'$. Taking the union over all pairs of clusters shows that $(p, p + b)$ is still disjoint from $2K^\sharp$. Hence $\text{gap}(2K^\sharp) \geq b$. Since $2K \subseteq 2K^\sharp$ and the two sumsets have the same endpoints, Lemma 2.1 gives $\text{gap}(2K^\sharp) \leq b$. Therefore $\text{gap}(2K^\sharp) = b$. $\square$

Corollary 2.3. *If Theorem 1.1 has a strict counterexample, then it has a strict counterexample of the form*

$$K = I_1 \cup \cdots \cup I_r,$$

where each I_i is a closed interval or a point, the intervals are in increasing order, and every component gap has length at least $b = \text{gap}(2K)$.

Proof. For a strict counterexample, $\delta = 2b - a - c > 0$, so $b > 0$. Apply Lemma 2.2. The completed set has the same diameter, the same value of b , and no larger values of a or c . Hence it remains a strict counterexample. By construction its component gaps are all at least b , and therefore the completed set is a finite union of intervals or points. $\square$

3. REDUCTION TO AT MOST FIVE COMPONENTS

We now prove the main structural reduction. Let

$$K = I_1 \cup \cdots \cup I_r, \quad I_i = [x_i, y_i],$$

where

$$0 = x_1 \leq y_1 < x_2 \leq y_2 < \cdots < x_r \leq y_r = D.$$

Write

$$w_i = y_i - x_i, \quad h_i = x_{i+1} - y_i \quad (1 \leq i < r).$$

Thus $w_i \geq 0$, and h_i is the gap between I_i and I_{i+1} .

Assume K is a completed strict counterexample, so

$$h_i \geq b \quad (1 \leq i < r).$$

Choose a largest gap $(p, p + b)$ of $2K$. Since $D \in 2K$, this gap cannot cross D . Replacing K by its reflection $D - K$ if necessary, we may assume

$$(3.1) \quad p + b \leq D.$$

Reflection does not change D, a, b, c .

Lemma 3.1 (Endpoint widths). *For a completed strict counterexample,*

$$2w_1 \geq b - c, \quad 2w_r \geq b - c.$$

Proof. The first two triple-sum intervals near the left endpoint are

$$3I_1 = [0, 3w_1]$$

and

$$2I_1 + I_2 = [x_2, 2y_1 + y_2].$$

Every other triple-sum interval has left endpoint at least x_2 . Therefore, if these two intervals are separated, the corresponding gap in $3K$ has length

$$x_2 - 3w_1 = (w_1 + h_1) - 3w_1 = h_1 - 2w_1.$$

Since $\text{gap}(3K) = c$, we have

$$h_1 - 2w_1 \leq c.$$

Using $h_1 \geq b$ gives $2w_1 \geq b - c$.

The proof at the right endpoint is identical. The last two triple-sum intervals are $I_{r-1} + 2I_r$ and $3I_r$, and their possible gap has length $h_{r-1} - 2w_r$. Hence $h_{r-1} - 2w_r \leq c$, and $h_{r-1} \geq b$ gives $2w_r \geq b - c$. $\square$

Lemma 3.2 (A row crossing). *For a completed strict counterexample satisfying (3.1),*

$$a \geq b + w_1.$$

Consequently,

$$b - c \geq 2\delta,$$

where $\delta = 2b - a - c$.

Proof. Consider the row of pair-sum intervals

$$I_1 + I_j \quad (1 \leq j \leq r).$$

The first interval $I_1 + I_1$ lies below the selected gap $(p, p + b)$, because it contains 0 and cannot meet the open gap. The last interval $I_1 + I_r$ lies above the selected gap: its right endpoint is $w_1 + D \geq D \geq p + b$, so it cannot lie below the gap and cannot cross it.

Therefore, as j increases, there is an index $j < r$ such that

$$I_1 + I_j \subseteq (-\infty, p] \quad \text{and} \quad I_1 + I_{j+1} \subseteq [p + b, \infty).$$

Equivalently,

$$w_1 + y_j \leq p, \quad x_{j+1} \geq p + b.$$

Subtracting gives

$$h_j - w_1 = x_{j+1} - (w_1 + y_j) \geq b.$$

Thus $h_j \geq b + w_1$, and since a is the largest component gap of K , $a \geq b + w_1$.

By Lemma 3.1, $w_1 \geq (b - c)/2$. Hence

$$a \geq b + \frac{b - c}{2} = \frac{3b - c}{2}.$$

It follows that

$$\delta = 2b - a - c \leq 2b - \frac{3b - c}{2} - c = \frac{b - c}{2},$$

which is $b - c \geq 2\delta$. $\square$

Proposition 3.3. *Every completed strict counterexample has at most five components.*

Proof. For a strict counterexample, $D < 12\delta$. By Lemma 3.2,

$$b \geq b - c \geq 2\delta.$$

By Lemma 3.1,

$$w_1 \geq \delta, \quad w_r \geq \delta.$$

If $r \geq 6$, then there are at least five component gaps, each of length at least b . Therefore

$$D \geq w_1 + w_r + 5b \geq \delta + \delta + 5(2\delta) = 12\delta,$$

contradicting $D < 12\delta$. $\square$

Thus a strict counterexample must be a completed union of r intervals or points with $2 \leq r \leq 5$. The rest of the proof eliminates these cases.

4. THE FINITE INTERVAL PROBLEM

In this section we formulate the finite problem that remains after Proposition 3.3. Scale the set by $1/\delta$. We may therefore assume

$$(4.1) \quad \delta = 2b - a - c = 1.$$

A strict counterexample would satisfy

$$(4.2) \quad D < 12.$$

Moreover

$$(4.3) \quad a = 2b - c - 1.$$

For $1 \leq i \leq r$, set

$$x_i = \sum_{k < i} (w_k + h_k), \quad y_i = x_i + w_i,$$

so $I_i = [x_i, y_i]$ and

$$D = \sum_{i=1}^r w_i + \sum_{i=1}^{r-1} h_i.$$

The variables of the finite problem are

$$w_1, \dots, w_r, h_1, \dots, h_{r-1}, b, c, p,$$

all constrained to be nonnegative.

4.1. The selected pair-sum gap. Let

$$P_r = \{(i, j) : 1 \leq i \leq j \leq r\}.$$

For $\alpha = (i, j) \in P_r$, write

$$J_\alpha^{(2)} = I_i + I_j = [x_i + x_j, y_i + y_j].$$

The selected gap $(p, p + b)$ of $2K$ assigns every $J_\alpha^{(2)}$ to one side of the gap. Let

$$L = \{\alpha \in P_r : J_\alpha^{(2)} \subseteq (-\infty, p]\}.$$

Then L is a downset in the partial order

$$(i, j) \leq (k, \ell) \iff i \leq k \text{ and } j \leq \ell.$$

Indeed, if $(i, j) \in L$ and $(k, \ell) \leq (i, j)$, then $y_k + y_\ell \leq y_i + y_j \leq p$, so $(k, \ell) \in L$. It contains $(1, 1)$ and, because of the normalization $p + b \leq D$, it does not contain $(1, r)$. Conversely, for the finite check we consider every such downset L . The selected-gap constraints are

$$(4.4) \quad y_i + y_j \leq p \quad ((i, j) \in L),$$

$$(4.5) \quad x_i + x_j \geq p + b \quad ((i, j) \notin L).$$

The numbers of possible downsets are

r	2	3	4	5
$\#L$	1	3	7	15.

For each r and L , we also choose an index $m \in \{1, \dots, r - 1\}$ at which the largest component gap is attained:

$$(4.6) \quad h_m = a = 2b - c - 1.$$

The remaining component gaps obey

$$(4.7) \quad b \leq h_i \leq 2b - c - 1.$$

By Lemmas 3.1 and 3.2, every strict counterexample also obeys

$$(4.8) \quad b - c \geq 2,$$

$$(4.9) \quad 2w_1 \geq b - c,$$

$$(4.10) \quad 2w_r \geq b - c,$$

$$(4.11) \quad p + b \leq D.$$

Remark 4.1. The finite check below does not need to impose the full condition $\text{gap}(2K) \leq b$. The selected-gap constraints (4.4)–(4.5) imply that $2K$ avoids the open interval $(p, p + b)$ and has points on both sides of it, so some gap of $2K$ has length at least b . Omitting the other pair-sum gap constraints only enlarges the feasible region. Therefore, if the enlarged system has no point with $D < 12$, then no strict counterexample exists.

4.2. Triple-sum cuts. Let

$$T_r = \{(i, j, k) : 1 \leq i \leq j \leq k \leq r\}.$$

For $\tau = (i, j, k) \in T_r$, write

$$J_\tau^{(3)} = I_i + I_j + I_k = [s_\tau, e_\tau].$$

Then

$$3K = \bigcup_{\tau \in T_r} J_\tau^{(3)}.$$

The condition $\text{gap}(3K) \leq c$ is encoded by cuts.

Lemma 4.2 (Cut criterion). *Let $\mathcal{J} = \{[s_\tau, e_\tau] : \tau \in T\}$ be a finite family of compact intervals, and let $t \geq 0$. Then*

$$\text{gap}\left(\bigcup_{\tau \in T} [s_\tau, e_\tau]\right) \leq t$$

if and only if, for every nonempty proper subset $S \subset T$, there exist $\sigma \in S$ and $\tau \in T \setminus S$ such that

$$(4.12) \quad s_\tau - e_\sigma \leq t.$$

Proof. If the union has a gap (u, v) of length $> t$, let S be the set of intervals lying on the left side of the gap. Then for every $\sigma \in S$ and every $\tau \notin S$ one has $e_\sigma \leq u$ and $s_\tau \geq v$, so $s_\tau - e_\sigma > t$. Thus (4.12) fails for this cut.

Conversely, suppose (4.12) fails for a nonempty proper S . Then

$$s_\tau > e_\sigma + t$$

for every $\sigma \in S$ and $\tau \notin S$. Let

$$M = \max_{\sigma \in S} e_\sigma, \quad N = \min_{\tau \in T \setminus S} s_\tau.$$

Choosing σ with $e_\sigma = M$ shows that $N > M + t$. No interval indexed by S extends past M , and no interval indexed by $T \setminus S$ begins before N . Thus (M, N) is a gap of the union of length greater than t . $\square$

In the finite check, every cut $S \subset T_r$ gives the disjunction

$$(4.13) \quad \bigvee_{\substack{\sigma \in S \\ \tau \in T_r \setminus S}} \{s_\tau - e_\sigma \leq c\}.$$

The certificate branches on such disjunctions. Each branch adds one linear inequality of the form $s_\tau - e_\sigma \leq c$.

5. THE RATIONAL CERTIFICATE

This section describes the exact certificate used to eliminate the finite problem of Section 4. The certificate file is

`sumset_gap_certificate.json`,

and the verifier is

`verify_sumset_gap_certificate.py`.

Their SHA-256 hashes are as follows:

`sumset_gap_certificate.json:`
 31ee47a60726ae8d64d75ac3c64d4474
 ea8c87ab4aa23a50f792e77698626af2

`verify_sumset_gap_certificate.py:`
 5c807d4c20638ce50168c2912e784ca8
 c7b9a2cdecb06bbb5bd3badebecd6a0de

The verifier uses Python's `fractions.Fraction` class and no numerical linear-programming solver.

5.1. Root systems. For each

$$2 \leq r \leq 5,$$

for each allowable downset $L \subset P_r$, and for each index $m \in \{1, \dots, r-1\}$ at which the largest component gap is attained, the verifier constructs the linear system consisting of:

- (i) nonnegativity of all variables;
- (ii) the completed-gap inequalities $h_i \geq b$;
- (iii) the upper bounds $h_i \leq 2b - c - 1$;
- (iv) the equality $h_m = 2b - c - 1$;
- (v) the derived inequalities $b - c \geq 2$, $2w_1 \geq b - c$, $2w_r \geq b - c$;
- (vi) the lower-half condition $p + b \leq D$;
- (vii) the selected pair-sum gap constraints (4.4)–(4.5).

There are

$$1 + 6 + 21 + 60 = 88$$

such root systems.

5.2. Branching. Starting from a root system, the verifier reads a finite rooted tree from the certificate. At every internal node the certificate specifies a cut $S \subset T_r$. By Lemma 4.2, any genuine counterexample satisfying $\text{gap}(3K) \leq c$ must satisfy at least one inequality

$$s_\tau - e_\sigma \leq c$$

with $\sigma \in S$ and $\tau \notin S$. The verifier checks that the children of that node are exactly all these alternatives. Thus every genuine counterexample follows at least one branch of the tree.

The leaves are of two kinds: infeasibility leaves and bound leaves. Each leaf is certified by a rational Farkas identity.

5.3. Farkas identities. At a leaf, the accumulated linear system has the form

$$Ax \leq B, \quad Ex = G, \quad x \geq 0,$$

where x is the vector of variables. A *bound leaf* stores rational multipliers

$$\lambda \geq 0, \quad \nu \geq 0, \quad \mu \text{ arbitrary},$$

such that the following identity of affine-linear functions holds exactly:

$$(5.1) \quad D(x) - 12 = \lambda \cdot (B - Ax) + \mu \cdot (G - Ex) + \nu \cdot x.$$

For every feasible x , the right-hand side is nonnegative; hence $D(x) \geq 12$.

An *infeasibility leaf* stores rational multipliers with the same sign conditions such that

$$(5.2) \quad -1 = \lambda \cdot (B - Ax) + \mu \cdot (G - Ex) + \nu \cdot x.$$

This is impossible for a feasible x , since the right-hand side would be non-negative. Hence such a leaf contains no feasible point.

The verifier checks (5.1) or (5.2) coefficient by coefficient over the rationals. It also checks that the root systems are exactly the 88 systems above and that every branch disjunction is complete.

Theorem 5.1 (Certified finite elimination). *There is no completed interval union with $2 \leq r \leq 5$, normalized by $\delta = 1$, satisfying the linear constraints forced by a strict counterexample, the triple-sum condition $\text{gap}(3K) \leq c$, and $D < 12$.*

Proof. The accompanying verifier checks the certificate exactly. It reconstructs every root system, checks branch completeness, and then checks every Farkas identity coefficient by coefficient. Its output is:

Verified 1073 rational Farkas leaves over 88 roots.
All internal branch disjunctions are complete.
No numerical LP solver was used.

The 1073 leaves consist of 969 infeasibility leaves and 104 bound leaves. By the branch-completeness check, every point satisfying the root constraints and $\text{gap}(3K) \leq c$ reaches a leaf. If it reaches an infeasibility leaf, it cannot exist. If it reaches a bound leaf, then $D \geq 12$ by (5.1). Thus no feasible point with $D < 12$ exists. $\square$

Remark 5.2. The generation of the certificate is irrelevant to the proof. Once the certificate is given, correctness reduces to exact rational arithmetic and to the elementary cut criterion of Lemma 4.2. The verifier reconstructs the linear systems from the mathematical definitions above; it does not trust any stored matrices from the certificate.

6. PROOF OF THE MAIN THEOREM

We now assemble the proof.

Proof of Theorem 1.1. Suppose, for contradiction, that there is a compact strict counterexample K . Normalize $\min K = 0$ and $\max K = D$. Let

$$a = \text{gap}(K), \quad b = \text{gap}(2K), \quad c = \text{gap}(3K),$$

and put

$$\delta = 2b - a - c.$$

The strict failure of the desired inequality is exactly

$$D < 12\delta,$$

so $\delta > 0$.

By Corollary 2.3, we may assume that K is a completed union of intervals or points whose component gaps are all at least b . By Proposition 3.3, this completed counterexample has at most five components. Since $\delta > 0$, it has at least two components.

Scale K by $1/\delta$. Then $\delta = 1$, and the strict counterexample condition becomes $D < 12$. Reflect K if necessary so that the selected largest gap of $2K$ lies in the lower half, $p + b \leq D$. Choose the downset L of pair-sum intervals lying below the selected gap, and choose an index m where the largest component gap of K is attained.

The resulting variables satisfy one of the 88 root systems described in Section 5.1. Since $c = \text{gap}(3K)$, they also satisfy the triple-sum cut disjunctions of Lemma 4.2. Hence Theorem 5.1 applies and gives $D \geq 12$, contradicting $D < 12$.

Therefore no strict counterexample exists. This proves

$$\text{gap}(K) + \text{gap}(3K) - 2\text{gap}(2K) \geq -\frac{\text{diam}(K)}{12}$$

for every compact $K \subset \mathbb{R}$. For finite sets, gap agrees with the usual largest consecutive gap g , so the finite-set statement follows. $\square$

Combining Theorem 1.1 with Proposition 1.2 gives the exact infimum.

Corollary 6.1. *For finite sets $A \subset \mathbb{R}$ with $|A| \geq 2$,*

$$\inf_A \frac{g(A) + g(3A) - 2g(2A)}{d(A)} = -\frac{1}{12}.$$

7. APPENDIX: RUNNING THE VERIFIER

Place the verifier and certificate in the same directory, or pass the certificate path explicitly. The verification command is

```
python3 verify_sumset_gap_certificate.py \
    sumset_gap_certificate.json
```

It should print

Verified 1073 rational Farkas leaves over 88 roots.

All internal branch disjunctions are complete.

No numerical LP solver was used.

The verifier performs the following checks.

- (1) It reconstructs the pair labels P_r , triple labels T_r , and all allowable downsets L for $2 \leq r \leq 5$.
- (2) It checks that the certificate has exactly the 88 required roots.
- (3) For each root, it reconstructs the base linear constraints from the formulas in Section 4.
- (4) At every internal node, it checks that the branch is a triple-sum cut and that the children are exactly all alternatives in the cut disjunction (4.13).
- (5) It validates labels, root indices, and sparse multiplier lists, rejecting duplicate multiplier indices.
- (6) At every leaf, it checks the relevant Farkas identity, (5.1) or (5.2), using exact rational arithmetic.

REFERENCES

- [1] Fabius Wiesner, *Lower bound for expression with biggest gaps in sumsets*, MathOverflow question 511261, 2026. Accessed May 14, 2026. <https://mathoverflow.net/questions/511261/lower-bound-for-expression-with-biggest-gaps-in-sumsets>